What a smartphone protocol can and cannot establish

Urban noise in Hanoi — three months, 363 measurements, three negative results

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Handover meeting

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Why noise in Hanoi

The gap this project sits in

- Dense, **motorcycle-dominated** traffic
- **No national strategic noise mapping programme**, no statutory noise action plan in Vietnam

  `Nguyen et al. 2025, City Env. Interactions`
- **One** instrumented campaign ever published for Hanoi — and it measured in **2005–2007**

  `Phan et al. 2010, Applied Acoustics`
- Class 1 meters and commercial models: out of reach for most local research budgets

The question

How far does a protocol built from **three consumer smartphones** and **open data** actually get you?

Research questions — and what is out of scope

1. Can OSM morphology predict measured noise at an **unmeasured location**?
2. Does a model trained in one city **transfer** to another?
3. Does vehicle **flow from video** explain measured levels?
4. What can a protocol claim with **no absolute reference**?

Out of scope — stated, not discovered later

Night-time noise (10 measurements after 21:00, none 00:00–05:00) · compliance assessment against any standard · any prediction beyond the three measured sites

State of the art — ten references, selected by use

What we lean on

- **Roberts et al. 2017**, *Ecography* — a CV block must exceed the autocorrelation range of the predictors
- **Meyer & Pebesma 2021**, *MEE* — area of applicability
- **Kardous & Shaw 2014**, *JASA* — smartphones depart from reference instruments by several dB, non-constant
- **Phan et al. 2010** — the Hanoi anchor

Source policy, applied

- `verified` — read in the source (12)
- `to_check` — second-hand, flagged (2)
- `grey` — press-reported, undocumented protocol **never a primary reference** (1)

The test of a reliability policy

The **most convenient** figures we had — twelve Hanoi arteries, via the press — are the ones classified `grey` and excluded.

Data

What we collected, and what we publish

Dataset	Size	Published	Licence
Field measurements	363 points, 3 sites	yes	CC-BY-4.0
Vehicle counts from video	147 videos	yes	CC-BY-4.0
Survey forms (XLSForm)	2 forms	yes	CC-BY-4.0
Traffic videos	147, 6.0 GB	no	not distributed
Raw Kobo exports	—	no	not distributed
OpenStreetMap extract	49 146 buildings	no	ODbL

Ethics — stated factually

No institutional ethics review was requested for the video collection. No approval, no exemption, no compliance is claimed. The originals contain faces and licence plates and were not anonymised at source; **only the aggregate counts are released**.

`docs/data-sources.md -- recommendation: submit to the VinUni ethics committee before any new campaign`

Method

The chain

acquisition	→	**preparation**	→	**features**	→	**selection**	→	**map + simulation**
ODK / Kobo	→	cleaning	→	OSM 300 m	→	8 models	→	40 m grid
363 points	→	+ weather	→	morphology	→	**3 CV protocols**	→	17 hours
	→		→		→	**code picks**	→	GAMA agents

Three protocols, one set of splits

Block-CV 600 m — fast, permissive · **Buffered LOO** 300 m — *the reference: its exclusion radius equals the feature radius* · **Leave-one-site-out** — an unseen typology

`docs/methodology.md -- 7 assumptions and 3 limitations, each stated where it is made`

Three months, and one pivot

June	Reproduce the Uganda dataset; plan: pretrain there, transfer to Hanoi, calibrate locally
June–July	Field campaign: 363 measurements, 3 typologies, 147 videos
July	First model, $R^2 = 0.45$ announced, GAMA simulation built
5 August	**Internal audit → pivot.** No sound level meter would become available; the campaign closed. The project stopped trying to produce a reference noise map — *it cannot* — and became a methodological study
5–6 August	Honest CV, baselines, ablation. $R^2 = 0.45$ withdrawn: the split was grouped on 110 m cells against a 300 m feature radius
11–12 August	Restructuring for publication and handover

Results

The ranking inverts — and that is the result

Model	**Block-CV 600 m** permissive	**Buffered LOO** **reference**	**Leave-one-site-out** unseen typology
Site × hour table	−0.008	−0.419	−0.058
log(dist_road), 2 param.	0.221	0.200	0.189
Physical kernel, 3 param.	0.255	0.246	0.222
LightGBM v1 (6 features)	0.304	0.137	0.029
LightGBM v2 (8 features)	0.332	0.099	−0.035
Hybrid (physics + ML residual)	**0.395**	0.123	0.035

The models that win the permissive split are the ones that lose the strict splits, **monotonically**. That is the signature of a model learning the sample's spatial autocorrelation, not physics.

`models/model_comparison.md -- 95% CI by block bootstrap, n = 363, 17 blocks`

The bias–variance trade-off, in numbers

In-sample fit *(worse)*	
R^2	0.499 → 0.166
MAE	3.79 → 5.30 dB
within ±5 dB	72.8 → 53.1%
σ simulated	5.57 → 3.14 dB

Generalisation *(better)*	
Buffered LOO	0.137 → 0.246
Leave-one-site-out	0.029 → 0.222

The three-parameter kernel fits the training points *less* closely and predicts unmeasured places *better*. The map is flatter because the model is less flexible.

`docs/methodology.md §5.3 -- MAE 5.30 dB read without this slide looks like a bad model`

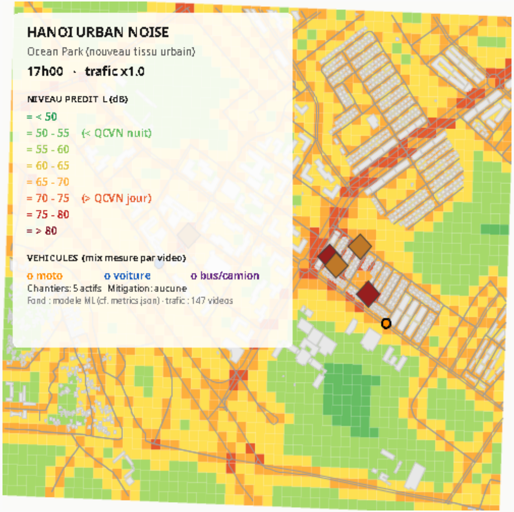
The delivered model, and the map

$$E = \frac{A_{hw}}{\max(d_{hw}, D_0)} + \frac{A_{res}}{\max(d_{res}, D_0)} + B$$

$$L = 10 \log_{10} E$$

A line-source attenuation law: energy falls as $1/d$ from two road classes, plus a background term. **Three parameters**, fitted on 363 points, all constrained non-negative. The learned residual is

computed at every run and **deliberately not applied**: under the reference protocol it degrades the prediction.



`Ocean Park, 17:00. 40 m grid, 5 587 cells`

What the restructuring found

We found one of our own published results was wrong

How

Translating figure labels. Re-running the script produced **a different file**.

The published validation validated **a grid that no longer existed**: regenerated on 5 August when the physical kernel replaced the hybrid, while the validation was last built six days earlier. **Nothing expressed that dependency.**

Published (wrong)
MAE 3.79 dB, r 0.715, 72.8% within $\pm$5 dB

Correct
MAE 5.30 dB, r 0.444, 53.1% within $\pm$5 dB

The old figures were **more flattering** and wrong. Published with the trade-off explanation; the old ones archived with their reason; the Makefile now carries the dependency so `make` rebuilds what is stale.

Verify by executing, not by reading

Every defect this project shipped was one that **reading could not catch**.

Guard (it runs)	What it catches
`test_cv_protocols`	the 110 m vs 300 m leak behind R^2=0.45
`test_grid_extent`	a map extended to unmeasured ground
`test_features_extraction`	drift in the input to every published number
`test_gama_fingerprint`	compares *values*, never strings
`test_pipeline_end_to_end`	a broken import, *including inside a function*
Makefile dependencies	a derived artefact older than its input
`make` environment check	an obscure `ModuleNotFoundError`

34 tests. Two habits: fingerprint before changing anything; compare commit dates across the whole tree.

Ten reasoning gaps, moved from code into documentation

Each was written only in a comment, where a reviewer would not look.

- **The kernel is fitted in dB, not in energy.** Levels span 47–88 dB — four orders of magnitude in energy — so a least-squares fit in energy would be driven by the loudest points alone. *A choice of loss function that determines the published coefficients.*
- **The construction source is calibrated on medians.** Means gave 74.7 dB, i.e. +8 dB simulated near a site, against +2 dB observed.
- **The exported fleet shares are shares of *density*, not of flow.** In congestion density is maximal while flow collapses.

On the calibration

A choice made not by preferring the statistic that behaves better, but by **rejecting the one whose output contradicts a field observation**.

Limits

What we cannot claim

- **The instrument is a smartphone application**, not a class 1 or 2 sound level meter — the class QCVN 26:2010 requires. This governs the uncertainty of every number in this talk.
- **Levels are relative, not absolute.** Three phones cross-calibrated against each other, never against a reference. A common bias is invisible by construction.
- **The vehicle detector is unvalidated.** Precision, recall and MAPE per class unknown; two-wheeler under-detection expected but unquantified. **Modal shares are not publishable.**
- **A 24 s window estimates its hour to roughly $\pm$30%** (1σ) from sampling alone, measured across 11 site–hour pairs.
- **The night is not sampled.** 10 measurements after 21:00.
- **$R^2 \approx 0.25$ is modest** — and it is the figure that survives a split excluding the feature support.

Handover

Where to start, ranked

1. **Validate the detector** — manual count on 10 videos, about one day. *Highest value per effort in the project*: it turns an open-ended limitation into a quantified uncertainty, and it is the input uncertainty of every downstream number.
2. **Extract `validation.py` and `physics.py`** from the 598-line evaluation script — the most valuable code is the least testable.
3. **A real propagation kernel: CNOSSOS-EU via NoiseModelling.** *Indicated by our own result*: if a two-parameter distance term beats a six-variable learned model, physics corrected by a local residual is the architecture the data points at.

Open items — placeholders, not guesses

Affiliation (CEI or COSMOS Lab) · one ORCID · a supervisor title · a co-authorship decision · **video retention: custodian, location, deletion deadline**

The repository is the deliverable.

`github.com/Colauz/noise-modelling-hanoi`

```
make setup && make features && make models && make results
```

Reproduces the maps and the model comparison from the two datasets that ship with it — no raw Kobo export, no 6 GB of video.